



DALLAS CRIME: POLICE INCIDENTS & ARRESTS

Students 1 and 2





INTRO





DATA SET

Dallas Police Incidents
(1,192,679 obs; 86 vars)

Dallas Police Arrests
(61,656 obs; 18 vars)

Crimes committed in Dallas between 2016-2019
(332,935 incidents; 29,100 arrests)





MOTIVE

To help law enforcement make Dallas a safer place and increase the awareness of residents by identifying criminal patterns and problematic areas in the city.





ANALYSIS

TABLE OF CONTENTS

01

WHAT

type of crime?

02

WHEN

do crimes occur?

03

WHERE

is improvement needed?

04

WHO

typically commits crime?

Crime Category Broken Down by Crime Subcategory



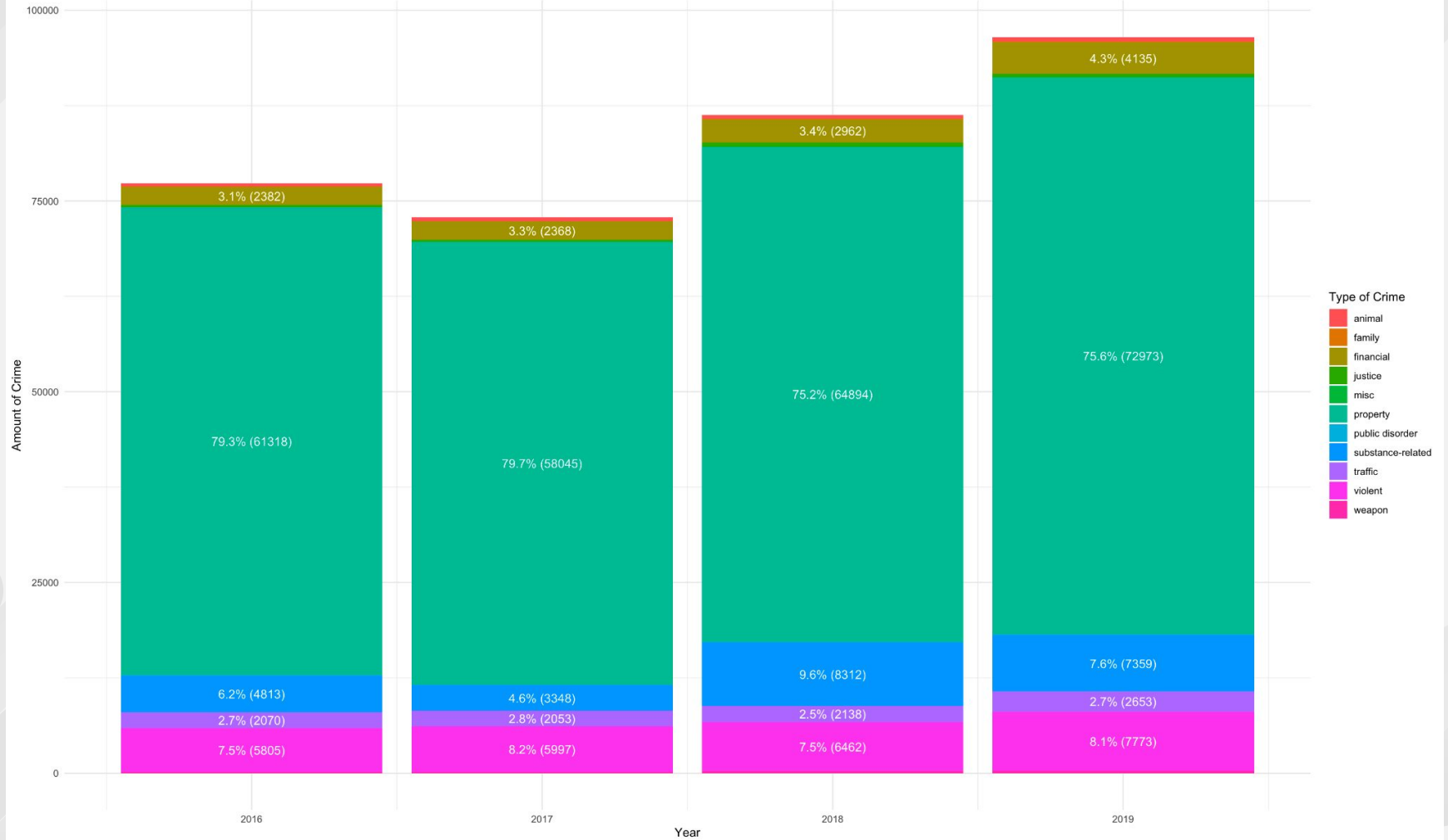
Crime Subcategory

- assault
- burglary
- drug-related
- fraud
- harassment
- intoxication
- murder
- robbery
- theft
- vandalism
- misc

Category	Count	Subcategory	Count
property	257230	burglary	87629
violent	26037	theft	85831
substance-related	23832	misc	48979
financial	11847	vandalism	41581
traffic	8914	intoxication	18973
animal	1949	assault	18562
justice	1662	robbery	14628
weapon	798	harassment	7195
family	394	fraud	5513
public disorder	253	drug-related	3636
misc	19	murder	408
Total			332935

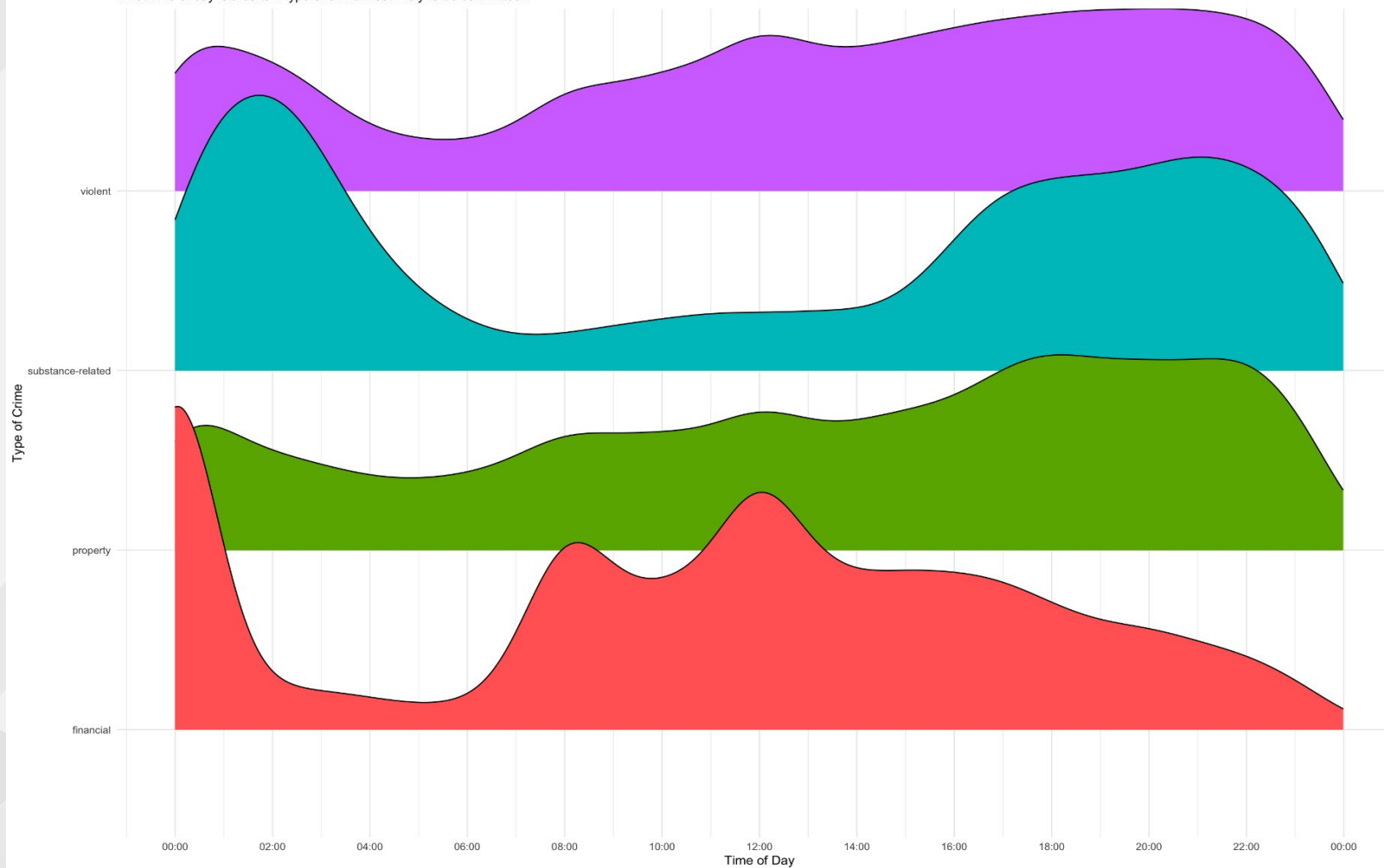
Criminal Activity Over Time

How has the frequency of crime type changed over time?



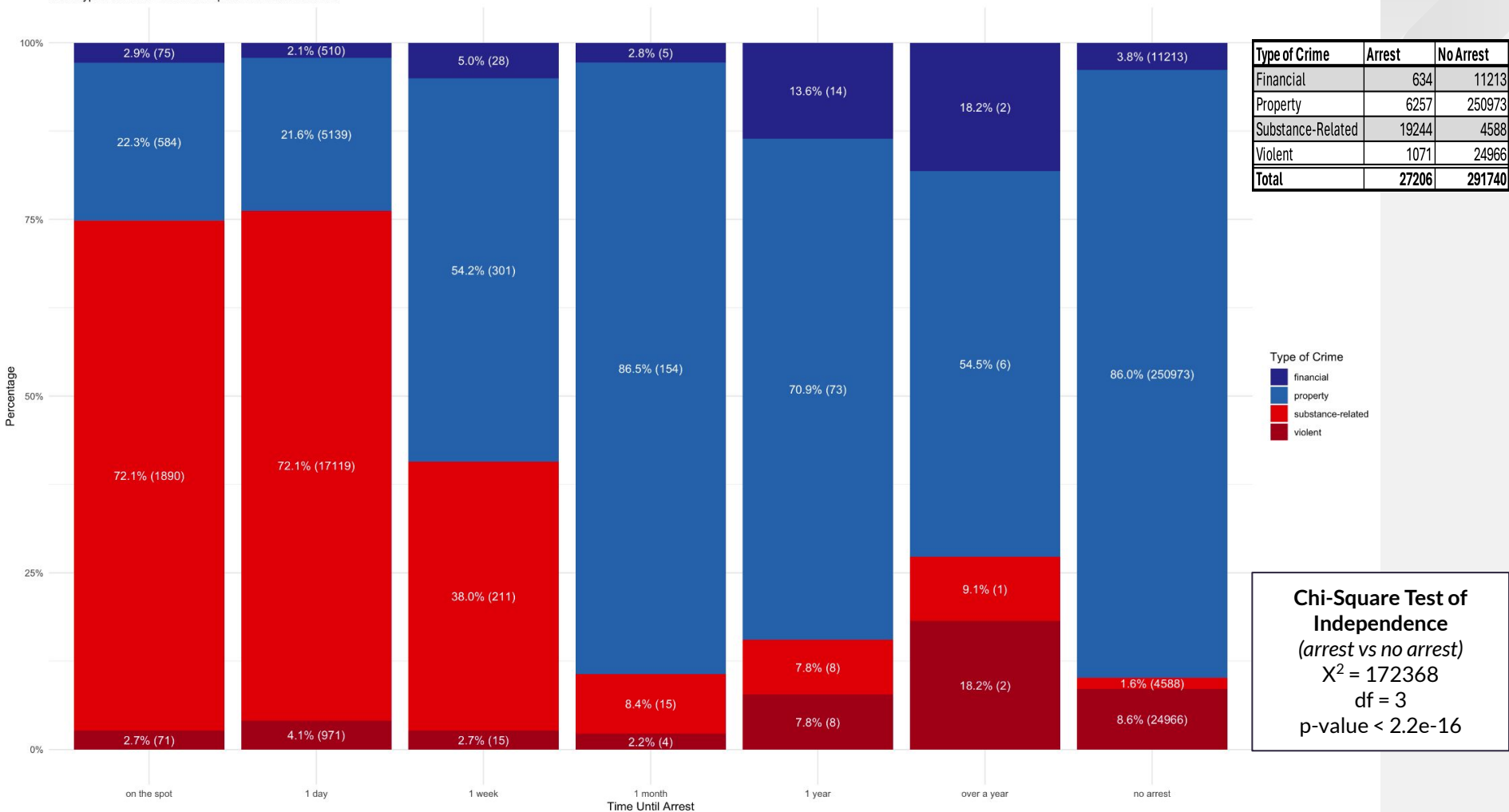
Distribution of Crime by Time of Day

What time of day is a certain type of crime most likely to be committed?

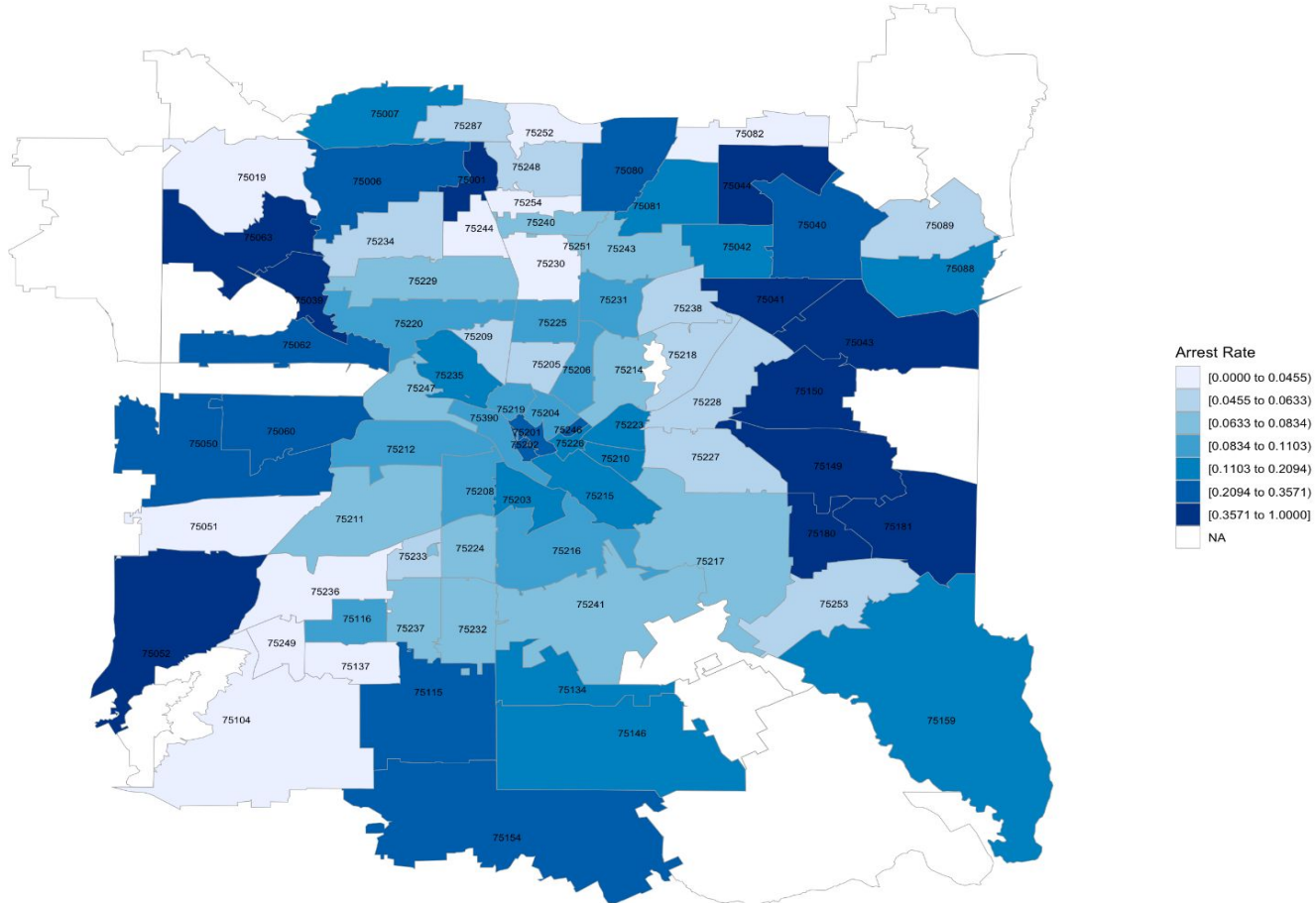


Time Until Arrest vs Type of Crime

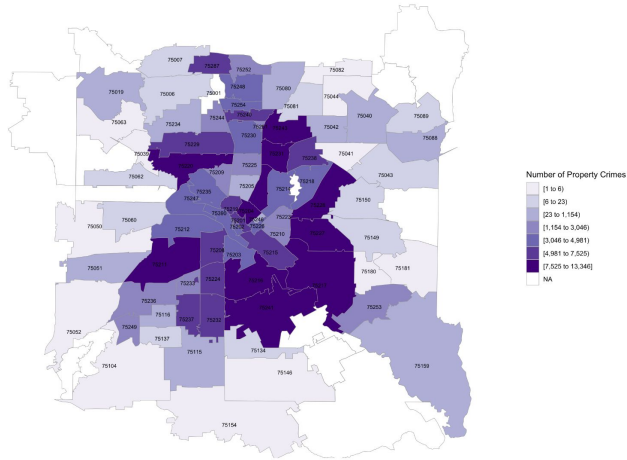
What types of crime lead to the quickest/slowest arrest?



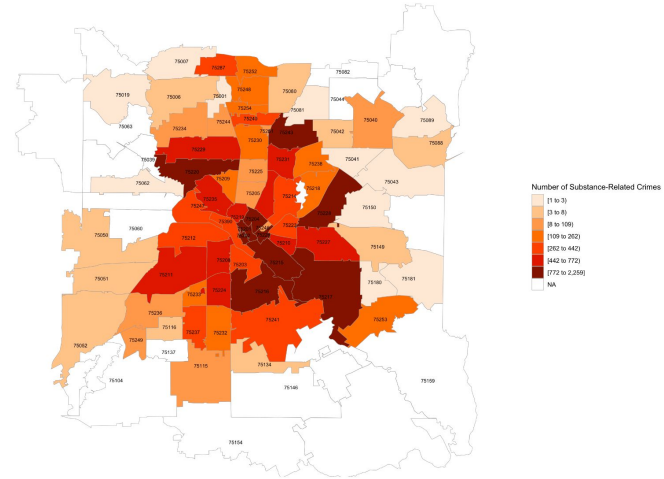
Arrest Rate by Zip Code in Dallas County



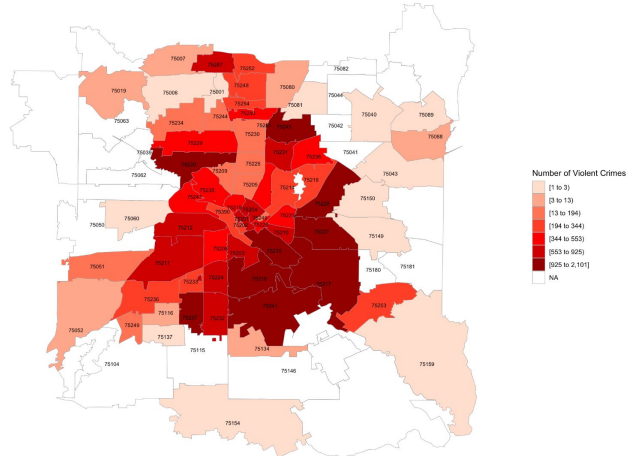
Number of Property Crimes by Zip Code in Dallas County



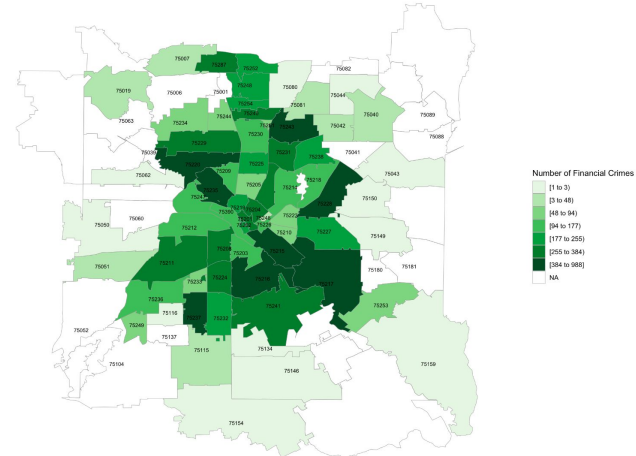
Number of Substance-Related Crimes by Zip Code in Dallas County



Number of Violent Crimes by Zip Code in Dallas County

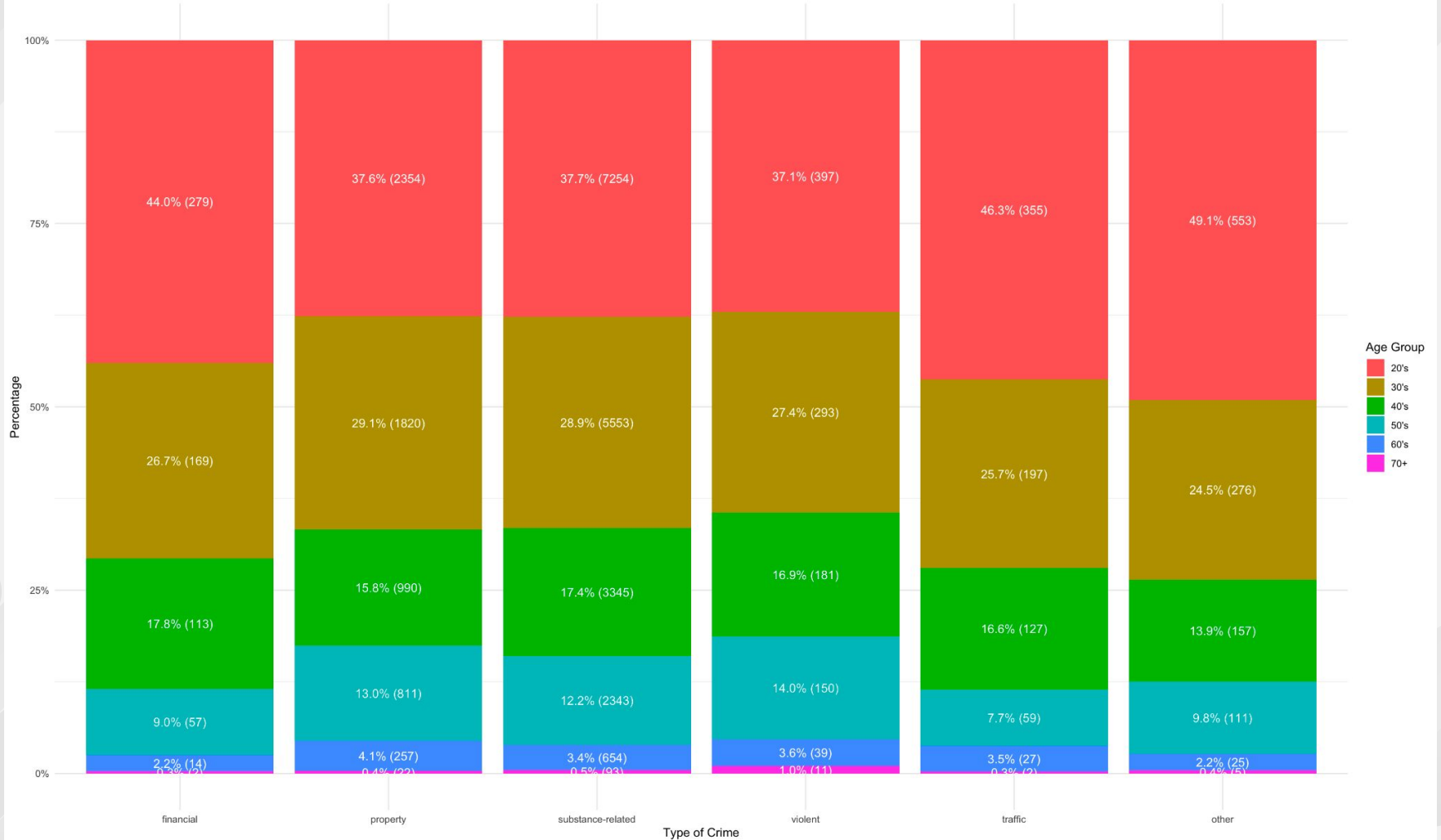


Number of Financial Crimes by Zip Code in Dallas County



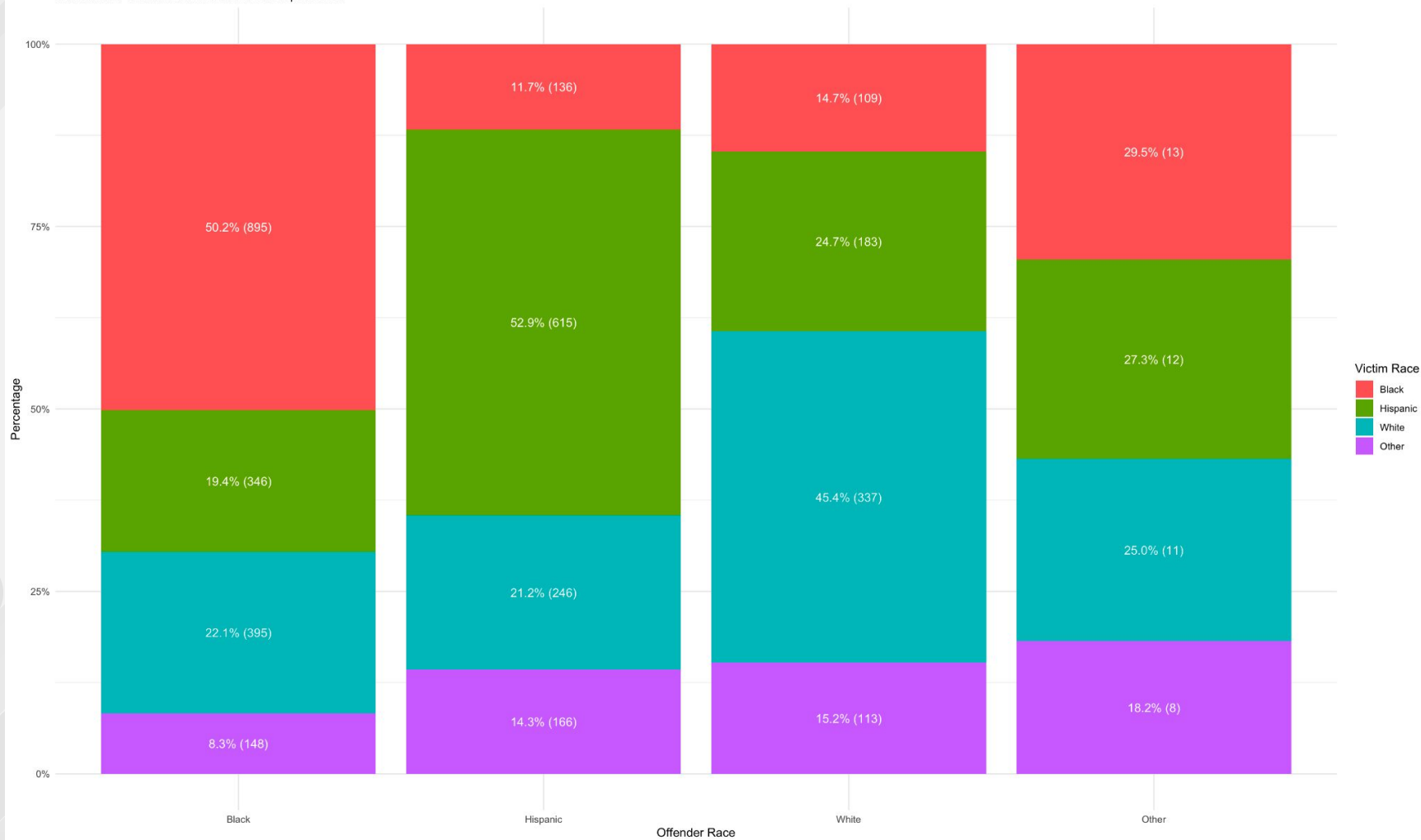
Type of Crime vs Age Group

Which age group is most likely to commit which type of crime?



Victim-Offender Combination

What kind of victim-offender combo is most prevalent?



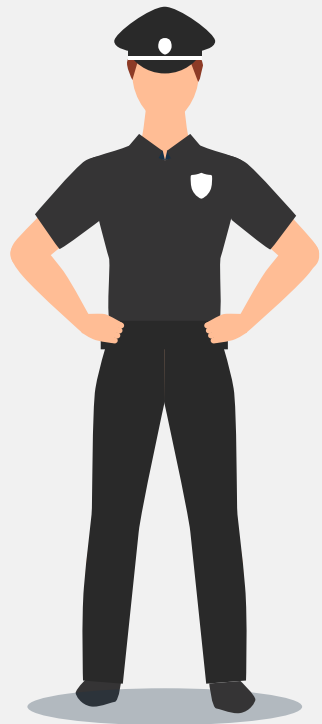


Just the Stats

Violent Victimization by Race or Hispanic Origin, 2008–2021

Table 2. Number of violent incidents, by victim and offender race/Hispanic origin, 2017–21

Victim race/ Hispanic origin	Total	Offender race/Hispanic origin					
		White ^{*a}	Black ^a	Hispanic ^b	Asian/Native Hawaiian/ Other Pacific Islander ^{a,c}	Other ^{a,d}	Unknown ^e
White ^a	15,795,650	8,721,450	2,382,400†	1,535,050†	187,150†	513,140†	1,821,890†
Black ^a	3,095,610	371,540	1,884,250†	243,880‡	11,850!	86,100†	341,240†
Hispanic	4,057,480	1,006,400	741,600†	1,423,520†	55,410†	65,770†	548,520†
Asian/Native Hawaiian/ Other Pacific Islander ^{a,c}	829,570	219,520	191,970	57,530†	113,220†	5,760!	122,640!
Other ^{a,d}	1,412,090	681,920	234,750†	127,740†	27,830†	178,530†	139,760†



CONCLUSION



FUTURE WORK

LIMIT 1

POPULATION

Take into account the size, demographics, and other socioeconomic factors of the population

LIMIT 2

POLICE EFFICIENCY

Assess the efficiency of the police force in certain beats / sectors to identify specific areas of improvement

LIMIT 3

COMPARISON

Compare our findings to other cities in the US: Houston, Los Angeles, Chicago

LIMIT 4

PREDICTIVE MODELING

Create a statistical model to predict crime and arrest success based on population demographics and type of crime

SUMMARY

How can past records on Dallas crime arrests and incidents be used to help understand areas of improvement for Dallas Police and criminal behavior?

The City of Dallas could benefit by...

- Improving resolution of property crime.
- Avoiding going out between 10pm and 2am.
- Increasing police presence in areas surrounding downtown.
- Implementing community outreach to address same-race violence.



THANKS!

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